

Definition : Subspaces of $\mathbb{R}^n$

A set U of vectors in $\mathbb{R}^n$ is called a subspace of $\mathbb{R}^n$ if it satisfies the following properties:

1. The zero vector $\mathbf{0} \in U$
2. If $\mathbf{x} \in U$ and $\mathbf{y} \in U$, then $\mathbf{x} + \mathbf{y} \in U$
3. If $\mathbf{x} \in U$, then $a\mathbf{x} \in U$ for all $a \in \mathbb{R}$

Definition : Null Space of a Matrix

The null space of an $m \times n$ matrix A is the set of all vectors for which multiplying by A gives 0. This is denoted:

$$\text{null } A = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$$

Definition : Image Space of a Matrix

The image space of an $m \times n$ matrix A is the set of all vectors which can be obtained by multiplying A by a vector in $\mathbb{R}^n$. This is denoted:

$$\text{im } A = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\}$$

Definition : Spanning Set

The set of all such linear combinations of a set of vectors $\mathbf{x}_i$ is called the span, and is denoted:

$$\text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\} = \{t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \dots + t_k\mathbf{x}_k \mid t_i \in \mathbb{R}\}$$

If $V = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$, we say V is spanned by the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$, or that the vectors span the space V .

Theorem : Span Theorem

Let $U = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ where each $\mathbf{x}_i \in \mathbb{R}^n$, then

1. U is a subspace of $\mathbb{R}^n$ containing each $\mathbf{x}_i$
2. If W is a subspace of $\mathbb{R}^n$ and each $\mathbf{x}_i \in W$, then $U \subseteq W$

Definition : Linear Independence in $\mathbb{R}^n$

We call a set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ of vectors linearly independent if:

$$t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \dots + t_k\mathbf{x}_k = \mathbf{0} \implies t_1 = t_2 = \dots = t_k = 0$$

Theorem

If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is an independent set of vectors in $\mathbb{R}^n$, then every vector in $\text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ has a unique representation as a linear combination of the $\mathbf{x}_i$.

Theorem : Independence Test

To verify that a set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ of vectors in $\mathbb{R}^n$ is independent, proceed as follows:

1. Set a linear combination equal to zero: $t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \dots + t_k\mathbf{x}_k = \mathbf{0}$
2. Show that $t_i = 0$ for each i

If some nontrivial linear combination vanishes, the vectors are not independent.

Theorem : Inverse Theorem

The following conditions are equivalent for an $n \times n$ matrix A :

1. A is invertible
2. $A\mathbf{x} = \mathbf{0} \implies \mathbf{x} = \mathbf{0}$ for $\mathbf{x} \in \mathbb{R}^n$
3. $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ for every $\mathbf{b} \in \mathbb{R}^n$

Theorem

Let A be an $m \times n$ matrix and let $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ denote the columns of A

1. $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ is independent in $\mathbb{R}^m$ iff $A\mathbf{x} = \mathbf{0}$ ($\mathbf{x} \in \mathbb{R}^n$) implies $\mathbf{x} = \mathbf{0}$
2. $\mathbb{R}^m = \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ iff $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ for every $\mathbf{b} \in \mathbb{R}^m$

Theorem

The following are equivalent for an $n \times n$ matrix A :

1. A is invertible
2. The columns of A are linearly independent
3. The columns of A span $\mathbb{R}^n$
4. The rows of A are linearly independent
5. The rows of A span the set of all $1 \times n$ rows

Theorem : Fundamental Theorem

Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

Definition : Basis of $\mathbb{R}^n$

If U is a subspace of $\mathbb{R}^n$, a set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ of vectors in U is called a basis of U if it satisfies the two following conditions:

1. $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ is linearly independent
2. $U = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$

Theorem : Invariance Theorem in $\mathbb{R}^n$

If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ and $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k\}$ are both bases of a subspace U of $\mathbb{R}^n$, then $m = k$.

Definition : Dimension of a Subspace of $\mathbb{R}^n$

If U is a subspace of $\mathbb{R}^n$ and $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ is any basis of U , the number m of vectors in the basis is called the dimension of U . This is denoted by:

$$\dim U = m$$

Note

We define:

$$\dim \{\mathbf{0}\} = 0$$

This means that $\{\mathbf{0}\}$ has a basis containing no vectors. This makes sense since $\mathbf{0}$ cannot belong to any independent set.

Theorem

Let $U \neq \{\mathbf{0}\}$ be a subspace of $\mathbb{R}^n$. Then,

1. U has a basis, and $\dim U \leq n$
2. Any independent set in U can be enlarged (by adding vectors from any fixed basis of U) to a basis of U
3. Any spanning set for U can be cut down (by deleting vectors) to a basis of U

Theorem

Let U be a subspace of $\mathbb{R}^n$ where $\dim U = m$ and let $B = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ be a set of m vectors in U . Then, B is independent iff B spans U .

Theorem

Let $U \subseteq W$ be subspaces of $\mathbb{R}^n$. Then,

1. $\dim U \leq \dim W$
2. If $\dim U = \dim W$, then $U = W$

Corollary

It follows from the previous theorem that if U is a subspace of $\mathbb{R}^n$, then $\dim U \in \{0, 1, 2, \dots, n\}$ and

$$\dim U = 0 \iff U = \{\mathbf{0}\}$$

$$\dim U = n \iff U = \mathbb{R}^n$$

The other subspaces of $\mathbb{R}^n$ are called proper.

Definition : Length in $\mathbb{R}^n$

The length of a vector $\mathbf{x}$, denoted $\|\mathbf{x}\|$, is defined by:

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Theorem

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$, then:

1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$
2. $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$
3. $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y}) \quad \forall a \in \mathbb{R}$
4. $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$
5. $\|\mathbf{x}\| \geq 0$, and $\|\mathbf{x}\| = 0$ iff $\mathbf{x} = \mathbf{0}$
6. $\|a\mathbf{x}\| = |a|\|\mathbf{x}\| \quad \forall a \in \mathbb{R}$

Theorem : Cauchy Inequality

If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|$$

Moreover, $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\|\|\mathbf{y}\|$ iff one of $\mathbf{x}$ and $\mathbf{y}$ is a multiple of the other.

The Cauchy inequality is equivalent to $(\mathbf{x} \cdot \mathbf{y})^2 \leq \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$

Corollary : Triangle Inequality

If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

Definition : Distance in $\mathbb{R}^n$

If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we define the distance function $d(\mathbf{x}, \mathbf{y})$ between $\mathbf{x}$ and $\mathbf{y}$ by

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

Theorem

If $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$, we have:

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$
2. $d(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = \mathbf{y}$
3. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$
4. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$

Definition : Orthogonal and Orthonormal Sets in $\mathbb{R}^n$

We say that two vectors $\mathbf{x}$ and $\mathbf{y} \in \mathbb{R}^n$ are orthogonal if $\mathbf{x} \cdot \mathbf{y} = 0$. More generally, a set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ of vectors in $\mathbb{R}^n$ is called an orthogonal set if

$$\mathbf{x}_i \cdot \mathbf{x}_j = 0 \quad \forall i \neq j \quad \text{and} \quad \mathbf{x}_i \neq \mathbf{0} \quad \forall i$$

Note that $\{\mathbf{x}\}$ is an orthogonal set if $\mathbf{x} \neq \mathbf{0}$. A set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ of vectors in $\mathbb{R}^n$ is called orthonormal if it is orthogonal and each $\mathbf{x}_i$ is a unit vector.

Theorem : Normalizing an Orthogonal Set

If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is an orthogonal set, then $\left\{ \frac{1}{\|\mathbf{x}_1\|} \mathbf{x}_1, \frac{1}{\|\mathbf{x}_2\|} \mathbf{x}_2, \dots, \frac{1}{\|\mathbf{x}_k\|} \mathbf{x}_k \right\}$ is an orthonormal set, and we say that is the result of normalizing the orthogonal set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$.

Theorem : Pythagoras' Theorem in $\mathbb{R}^n$

If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is an orthogonal set in $\mathbb{R}^n$, then

$$\|\mathbf{x}_1 + \mathbf{x}_2 + \dots + \mathbf{x}_k\|^2 = \|\mathbf{x}_1\|^2 + \|\mathbf{x}_2\|^2 + \dots + \|\mathbf{x}_k\|^2$$

Theorem : Expansion Theorem in $\mathbb{R}^n$

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be an orthogonal basis of a subspace U of $\mathbb{R}^n$. If $\mathbf{x}$ is any vector in U , we have

$$\mathbf{x} = \left(\frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \right) \mathbf{f}_1 + \left(\frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \right) \mathbf{f}_2 + \dots + \left(\frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \right) \mathbf{f}_m$$

The expansion of $\mathbf{x}$ as a linear combination of the orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is called the Fourier expansion of $\mathbf{x}$ and the coefficients $t_i = \frac{\mathbf{x} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$ are called the Fourier coefficients.

Definition : Column and Row Space of a Matrix

The column space, denoted $\text{col } A$, of A is the subspace of $\mathbb{R}^m$ spanned by the columns of A .

The row space, denoted $\text{row } A$, of A is the subspace of $\mathbb{R}^n$ spanned by the rows of A .

Lemma

Let A and B denote $m \times n$ matrices.

1. If $A \rightarrow B$ by elementary row operations, then $\text{row } A = \text{row } B$
2. If $A \rightarrow B$ by elementary column operations, then $\text{col } A = \text{col } B$

Lemma

If R is a row-echelon matrix, then:

1. The nonzero rows of R are a basis of $\text{row } R$
2. The columns of R containing leading ones are a basis of $\text{col } R$

Theorem : Rank Theorem

Let A denote any $m \times n$ matrix of rank r . Then,

$$\dim(\text{col } A) = \dim(\text{row } A) = r$$

Moreover, if A is carried to a row-echelon matrix R by row operations, then

1. The r nonzero rows of R are a basis of $\text{row } A$
2. If the leading 1s lie in columns $j_1, j_2, \dots, j_r$ of R , then columns $j_1, j_2, \dots, j_r$ of A are a basis of $\text{col } A$

Corollary

If A is any matrix, then $\text{rank } A = \text{rank } (A^T)$.

Corollary

If A is an $m \times n$ matrix, then $\text{rank } A \leq m$ and $\text{rank } A \leq n$.

Corollary

If A is an $m \times n$ matrix, U is an $m \times m$ invertible matrix, and V is an $n \times n$ invertible matrix, then:

$$\text{rank } A = \text{rank } (UA) = \text{rank } (AV)$$

Lemma

Let A, U, V be matrices of sizes $m \times n$, $p \times m$, and $n \times q$ respectively.

1. $\text{col}(AV) \subseteq \text{col } A$, with equality if $VV' = I_n$ for some V'
2. $\text{row}(UA) \subseteq \text{row } A$, with equality if $U'U = I_m$ for some U'

Corollary

If A is $m \times n$ and B is $n \times m$, then $\text{rank}(AB) \leq \text{rank } A$ and $\text{rank}(AB) \leq \text{rank } B$.

Theorem

Let A denote an $m \times n$ matrix of rank r . Then,

1. The $n - r$ basic solutions to the system $A\mathbf{x} = \mathbf{0}$ provided by the Gaussian algorithm are a basis of $\text{null } A$, so $\dim(\text{null } A) = n - r$
2. The rank theorem provides a basis of $\text{im } A = \text{col } A$, and $\dim(\text{im } A) = r$

Theorem

The following are equivalent for an $m \times n$ matrix A :

1. $\text{rank } A = n$
2. The rows of A span $\mathbb{R}^n$
3. The columns of A are linearly independent in $\mathbb{R}^m$
4. The $n \times n$ matrix $A^T A$ is invertible
5. $CA = I_n$ for some $n \times m$ matrix C
6. If $A\mathbf{x} = \mathbf{0}$, $\mathbf{x} \in \mathbb{R}^n$, then $\mathbf{x} = \mathbf{0}$

Theorem

The following are equivalent for an $m \times n$ matrix A :

1. $\text{rank } A = m$
2. The columns of A span $\mathbb{R}^m$
3. The rows of A are linearly independent in $\mathbb{R}^n$
4. The $m \times m$ matrix AA^T is invertible
5. $AC = I_m$ for some $n \times m$ matrix C
6. The system $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ in $\mathbb{R}^m$

Definition : Similar Matrices

If A and B are $n \times n$ matrices, we say that A and B are similar, and write $A \sim B$, if $B = P^{-1}AP$ for some invertible matrix P .

If $A \sim B$, then necessarily $B \sim A$. To see why, suppose that $B = P^{-1}AP$. Then $A = PBP^{-1} = Q^{-1}BQ$ where $Q = P^{-1}$ is invertible. This proves the second of the following properties of similarity:

1. $A \sim A$ for all square matrices A
2. If $A \sim B$, then $B \sim A$
3. If $A \sim B$ and $B \sim C$, then $A \sim C$

These properties say that the similarity relation $\sim$ is an equivalence relation on the set of $n \times n$ matrices.

Note

Similarity is compatible with inverses, transposes and powers:

$$\text{If } A \sim B \text{ then } A^{-1} \sim B^{-1}, \quad A^T \sim B^T, \quad \text{and} \quad A^k \sim B^k \quad \forall k \in \mathbb{Z}^+$$

Definition : Trace of a Matrix

The trace of an $n \times n$ matrix A , denoted $\text{tr } A$, is defined to be the sum of the main diagonal elements of A . That is, if $A = [a_{ij}]$, then $\text{tr } A = a_{11} + a_{22} + \cdots + a_{nn}$.

Lemma

Let A and B be $n \times n$ matrices. Then $\text{tr}(AB) = \text{tr}(BA)$.

Theorem

If A and B are similar $n \times n$ matrices, then A and B have the same determinant, rank, trace, characteristic polynomial, and eigenvalues.

Theorem

Let A be an $n \times n$ matrix.

1. The eigenvalues λ of A are the roots of the characteristic polynomial $c_A(x)$ of A
2. The λ -eigenvectors $\mathbf{x}$ are the nonzero solutions to the homogeneous system

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

Theorem

Let A be an $n \times n$ matrix.

1. A is diagonalizable iff $\mathbb{R}^n$ has a basis $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ consisting of eigenvectors of A
2. When this is the case, the matrix $P = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_n]$ is invertible and $P^{-1}AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ where, for each i , λ_i is the eigenvalue of A corresponding to $\mathbf{x}_i$

Theorem

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ be eigenvectors corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ of an $n \times n$ matrix A . Then $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a linearly independent set.

Theorem

If A is an $n \times n$ matrix with n distinct eigenvalues, then A is diagonalizable.

Lemma

Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ be a linearly independent set of eigenvectors of an $n \times n$ matrix A , extend it to a basis $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k, \dots, \mathbf{x}_n\}$ of $\mathbb{R}^n$, and let

$$P = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_n]$$

be the (invertible) $n \times n$ matrix with the $\mathbf{x}_i$ as its columns. If $\lambda_1, \lambda_2, \dots, \lambda_k$ are the (not necessarily distinct) eigenvalues of A corresponding to $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ respectively, then $P^{-1}AP$ has block form

$$P^{-1}AP = \begin{bmatrix} \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_k) & B \\ 0 & A_1 \end{bmatrix}$$

where B has size $k \times (n - k)$ and A_1 has size $(n - k) \times (n - k)$.

Note that this lemma with $k = n$ shows that an $n \times n$ matrix A is diagonalizable if $\mathbb{R}^n$ has a basis of eigenvectors of A .

Definition : Eigenspace of a Matrix

If λ is an eigenvalue of an $n \times n$ matrix A , define the eigenspace of A corresponding to λ by:

$$E_{\lambda}(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \lambda\mathbf{x}\}$$

Recall the multiplicity of an eigenvalue λ of A is the number of times λ occurs as a root of the characteristic polynomial $c_A(x)$ of A . In other words, the multiplicity of λ is the largest integer $m \geq 1$ such that

$$c_A(x) = (x - \lambda)^m g(x)$$

for some polynomial $g(x)$.

Lemma

Let λ be an eigenvalue of multiplicity m of a square matrix A . Then $\dim(E_{\lambda}(A)) \leq m$.

Note

When does $\dim(E_{\lambda}(A)) = m$ for each eigenvalue λ ? It turns out that this characterizes the diagonalizable $n \times n$ matrices A for which $c_A(x)$ factors completely over $\mathbb{R}$. By this we mean that $c_A(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n)$, where the λ_i are real numbers (not necessarily distinct). In other words, every eigenvalue of A is real.

Theorem

The following are equivalent for a square matrix A for which $c_A(x)$ factors completely.

1. A is diagonalizable
2. $\dim(E_{\lambda}(A))$ equals the multiplicity of λ for every eigenvalue λ of the matrix A

Definition : Vector Spaces

A vector space consists of a nonempty set V of objects (called vectors) that can be added, multiplied by a real number, and for which certain axioms hold. If $\mathbf{u}$ and $\mathbf{v}$ are two vectors in V , their sum is expressed as $\mathbf{u} + \mathbf{v}$, and the scalar product of $\mathbf{v}$ by a real number a is denoted as $a\mathbf{v}$. These axioms are called vector addition and scalar multiplication, respectively. The following axioms are assumed to hold.

Axioms for vector addition

- (i) If $\mathbf{u}$ and $\mathbf{v}$ are in V , then $\mathbf{u} + \mathbf{v}$ is in V
- (ii) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}$ and $\mathbf{v}$ in V
- (iii) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ for all $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ in V
- (iv) An element $\mathbf{0}$ in V exists such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$ for every $\mathbf{v}$ in V
- (v) For each $\mathbf{v}$ in V , an element $-\mathbf{v}$ in V exists such that $-\mathbf{v} + \mathbf{v} = \mathbf{0}$ and $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$

Axioms for scalar multiplication

- (i) If $\mathbf{v}$ is in V , then $a\mathbf{v}$ is in V for all a in $\mathbb{R}$
- (ii) $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$ for all $\mathbf{v}$ and $\mathbf{w}$ in V and all a in $\mathbb{R}$
- (iii) $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ for all $\mathbf{v}$ in V and all a and b in $\mathbb{R}$
- (iv) $a(b\mathbf{v}) = (ab)\mathbf{v}$ for all $\mathbf{v}$ in V and all a and b in $\mathbb{R}$
- (v) $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in V

Definition : Polynomials

The set $\mathbf{P}$ of all polynomials is a vector space with addition and scalar multiplication defined as follows for $p(x) = a_0 + a_1x + a_2x^2 + \dots$ and $q(x) = b_0 + b_1x + b_2x^2 + \dots$:

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots$$

$$ap(x) = aa_0 + (aa_1)x + (aa_2)x^2 + \dots$$

The zero vector is the zero polynomial, and the negative of a polynomial $p(x) = a_0 + a_1x + a_2x^2 + \dots$ is the polynomial $-p(x) = -a_0 - a_1x - a_2x^2 - \dots$ obtained by negating all the coefficients.

Definition : Polynomials of a Given Degree

Given $n \geq 1$, let $\mathbf{P}_n$ denote the set of all polynomials of degree at most n , together with the zero polynomial. That is

$$\mathbf{P}_n = \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_0, a_1, a_2, \dots, a_n \in \mathbb{R}\}$$

Then $\mathbf{P}_n$ is a vector space.

Definition : Functions over an Interval

The set $\mathbf{F}[a, b]$ of all functions on the interval $[a, b]$ is a vector space using point-wise addition $((f + g)(x) = f(x) + g(x) \quad \forall x \in [a, b])$ and scalar multiplication $((rf)(x) = rf(x) \quad \forall x \in [a, b])$. The zero function denoted 0 is the constant function defined by

$$0(x) = 0 \quad \forall x \in [a, b]$$

The negative of a function f is denoted $-f$ and has action defined by

$$(-f)(x) = -f(x) \quad \forall x \in [a, b]$$

Theorem : Cancellation

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in a vector space V . If $\mathbf{v} + \mathbf{u} = \mathbf{v} + \mathbf{w}$, then $\mathbf{u} = \mathbf{w}$.

Definition : Difference

We define the difference $\mathbf{u} - \mathbf{v}$ of two vectors in V as

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$$

Theorem

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in a vector space V , the equation

$$\mathbf{x} + \mathbf{v} = \mathbf{u}$$

has one and only one solution $\mathbf{x}$ in V given by

$$\mathbf{x} = \mathbf{u} - \mathbf{v}$$

Theorem

Let $\mathbf{v}$ denote a vector in a vector space V and let a denote a real number

- (i) $0\mathbf{v} = \mathbf{0}$
- (ii) $a\mathbf{0} = \mathbf{0}$
- (iii) If $a\mathbf{v} = \mathbf{0}$, then either $a = 0$ or $\mathbf{v} = \mathbf{0}$
- (iv) $(-1)\mathbf{v} = -\mathbf{v}$
- (v) $(-a)\mathbf{v} = -(a\mathbf{v}) = a(-\mathbf{v})$

Definition : Subspaces of a Vector Space

If V is a vector space, a nonempty subset $U \subseteq V$ is called a subspace of V if U is itself a vector space using the addition and scalar multiplication of V .

Theorem : Subspace Test

A subset U of a vector space is a subspace of V if and only if it satisfies the following three conditions:

- (i) $\mathbf{0}$ lies in U where $\mathbf{0}$ is the zero vector of V
- (ii) If $\mathbf{u}_1$ and $\mathbf{u}_2$ are in U , then $\mathbf{u}_1 + \mathbf{u}_2$ is also in U
- (iii) If $\mathbf{u}$ is in U , then $a\mathbf{u}$ is in U for each scalar a

Note

If V is any vector space, both $\{\mathbf{0}\}$ and V are subspaces of V . The vector space $\{\mathbf{0}\}$ is called the zero subspace of V .

Definition : $m \times n$ Matrices

The set $\mathbf{M}_{mn}$ of all $m \times n$ matrices is a vector space where addition and scalar multiplication are the standard definitions for matrices.

Definition : Evaluation

Suppose $p(x)$ is a polynomial and a is a number. Then the number $p(a)$ obtained by replacing x by a in the expression for $p(x)$ is called the evaluation of $p(x)$ at a . If $p(a) = 0$, the number a is called a root of $p(x)$.

Theorem

$\mathbf{P}_n$ is a subspace of $\mathbf{P}$ for each $n \geq 0$.

Theorem

The subset $\mathbf{D}[a, b]$ of all differentiable functions on $[a, b]$ is a subspace of the vector space $\mathbf{F}[a, b]$.

Definition : Linear Combinations and Spanning

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of vectors in a vector space V . As in $\mathbb{R}^n$, a vector $\mathbf{v}$ is called a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ if it can be expressed in the form

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$$

where $a_1, a_2, \dots, a_n$ are scalars, called the coefficients of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. The set of all linear combinations of these vectors is called their span, and is denoted by

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n \mid a_i \in \mathbb{R}\}$$

If it happens that $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, these vectors are called a spanning set for V .

Theorem

Let $U = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V . Then:

- (i) U is subspace of V containing each of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$
- (ii) U is the “smallest” subspace containing these vectors in the sense that any subspace that contains each of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ must contain U

Theorem : Linear Independence and Dependence

As in $\mathbb{R}^n$, a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is called linearly independent if it satisfies the following condition:

$$\text{If } s_1\mathbf{v}_1 + s_2\mathbf{v}_2 + \cdots + s_n\mathbf{v}_n = \mathbf{0}, \quad \text{then } s_1 = s_2 = \cdots = s_n = 0$$

Theorem

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent set of vectors in a vector space V . If a vector $\mathbf{v}$ has two representations:

$$\mathbf{v} = s_1\mathbf{v}_1 + s_2\mathbf{v}_2 + \cdots + s_n\mathbf{v}_n$$

$$\mathbf{v} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_n\mathbf{v}_n$$

as a linear combination of these vectors, then $s_1 = t_1, s_2 = t_2, \dots, s_n = t_n$. In other words, every vector in V can be written in a unique way as a linear combination of the $\mathbf{v}_i$.

Theorem : Fundamental Theorem

Suppose a vector space V can be spanned by n vectors. If any set of m vectors in V is linearly independent, then $m \leq n$.

Definition : Basis of a Vector Space

As in $\mathbb{R}^n$, a set $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ of vectors in a vector space V is called a basis of V if it satisfies the two following conditions:

- (i) $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is linearly independent
- (ii) $V = \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$

Theorem : Invariance Theorem

Let $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be two bases of a vector space V . Then $n = m$.

Definition : Dimension of a Vector Space

If $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis of the nonzero vector space V , the number n of vectors in the basis is called the dimension of V , and we write

$$\dim V = n$$

The zero vector space $\{\mathbf{0}\}$ is defined to have dimension 0:

$$\dim \{\mathbf{0}\} = 0$$

Note

The space $\mathbf{M}_{mn}$ has dimension mn , and one basis consists of all $m \times n$ matrices with exactly one entry equal to 1 and all other entries equal to 0. We call this the standard basis of $\mathbf{M}_{mn}$.

Note

The space $\mathbf{P}_n$ has dimension $n + 1$. The standard basis of this space is $\{1, x, x^2, \dots, x^n\}$.

Lemma : Independent Lemma

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an independent set of vectors in a vector space V . If $\mathbf{u} \in V$ but $\mathbf{u} \notin \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, then $\{\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is also independent.

Definition : Finite Dimensional and Infinite Dimensional Vector Spaces

A vector space V is called finite dimensional if it is spanned by a finite set of vectors. Otherwise, V is called infinite dimensional.

Note

The zero vector space $\{\mathbf{0}\}$ is finite dimensional because $\{\mathbf{0}\}$ is a spanning set.

Lemma

Let V be a finite dimensional vector space. If U is any subspace of V , then any independent subset of U can be enlarged to a finite basis of U .

Theorem

Let V be a finite dimensional vector space spanned by m vectors.

- (i) V has a finite basis, and $\dim V \leq m$
- (ii) Every independent set of vectors in V can be enlarged to a basis of V by adding vectors from any fixed basis of V
- (iii) If U is a subspace of V , then
 - (a) U is finite dimensional and $\dim U \leq \dim V$
 - (b) If $\dim U = \dim V$ then $U = V$

Theorem

Let U and W be subspaces of the finite dimensional space V

- (i) If $U \subseteq W$, then $\dim U \leq \dim W$
- (ii) If $U \subseteq W$ and $\dim U = \dim W$, then $U = W$

Lemma : Independent Lemma

A set $D = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of vectors in a vector space V is dependent if and only if some vector in D is a linear combination of the others.

Theorem

Let V be a finite dimensional vector space. Any spanning set for V can be cut down (by deleting vectors) to a basis of V .

Theorem

Let V be a vector space with $\dim V = n$, and suppose S is a set of exactly n vectors in V . Then S is independent if and only if S spans V .

Definition : Sum and Intersection

If U and W are subspaces of V , their sum $U + W$ and their intersection $U \cap W$ are:

$$U + W = \{\mathbf{u} + \mathbf{w} \mid \mathbf{u} \in U \text{ and } \mathbf{w} \in W\}$$

$$U \cap W = \{\mathbf{v} \mid \mathbf{v} \in U \text{ and } \mathbf{v} \in W\}$$

Theorem

Suppose that U and W are finite dimensional subspaces of a vector space V . Then $U + W$ is finite dimensional, and

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$$

Definition : Linear Transformations of Vector Spaces

If V and W are two vector spaces, a function $T : V \rightarrow W$ is called a linear transformation if it satisfies the following axioms.

- (i) $T(\mathbf{v} + \mathbf{v}_1) = T(\mathbf{v}) + T(\mathbf{v}_1) \quad \forall \mathbf{v}, \mathbf{v}_1 \in V$
- (ii) $T(r\mathbf{v}) = rT(\mathbf{v}) \quad \forall \mathbf{v} \in V, r \in \mathbb{R}$

Definition : Linear Operator

A linear transformation $T : V \rightarrow V$ is called a linear operator on V .

Note

If V and W are vector spaces, the following are linear transformations

- (i) Identity Operator $V \rightarrow V$

$$1_V : V \rightarrow V \text{ where } 1_V(\mathbf{v}) = \mathbf{v} \quad \forall \mathbf{v} \in V$$

- (ii) Zero Transformation $V \rightarrow W$

$$0 : V \rightarrow W \text{ where } 0(\mathbf{v}) = \mathbf{0} \quad \forall \mathbf{v} \in V$$

- (iii) Scalar Operator $V \rightarrow V$

$$a : V \rightarrow V \text{ where } a(\mathbf{v}) = a\mathbf{v} \quad \forall \mathbf{v} \in V, a \in \mathbb{R}$$

Theorem

Let $T : V \rightarrow W$ be a linear transformation

- (i) $T(\mathbf{0}) = \mathbf{0}$
- (ii) $T(-\mathbf{v}) = -T(\mathbf{v}) \quad \forall \mathbf{v} \in V$
- (iii) $T(r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \cdots + r_k\mathbf{v}_k) = r_1T(\mathbf{v}_1) + r_2T(\mathbf{v}_2) + \cdots + r_kT(\mathbf{v}_k) \quad \forall v_i \in V, r_i \in \mathbb{R}$

Definition : Equality of Linear Transformations

Two linear transformations $T : V \rightarrow W$ and $S : V \rightarrow W$ are called equal (written $T = S$) if they have the same action, that is, $T(\mathbf{v}) = S(\mathbf{v}) \quad \forall \mathbf{v} \in V$

Theorem

Let $T : V \rightarrow W$ and $S : V \rightarrow W$ be two linear transformations. Suppose that $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. If $T(\mathbf{v}_i) = S(\mathbf{v}_i)$ for each i , then $T = S$.

Theorem

Let V and W be vector spaces and let $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis of V . Given any vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$ in W (not necessarily distinct), there exists a unique linear transformation $T : V \rightarrow W$ satisfying $T(\mathbf{b}_i) = \mathbf{w}_i$, for each $i = 1, 2, \dots, n$. In fact, the action of T is as follows:

Given $\mathbf{v} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + \cdots + v_n\mathbf{b}_n \in V$, $v_i \in \mathbb{R}$, then

$$T(\mathbf{v}) = T(v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + \cdots + v_n\mathbf{b}_n) = v_1\mathbf{w}_1 + v_2\mathbf{w}_2 + \cdots + v_n\mathbf{w}_n$$

Definition : Kernel and Image of a Linear Transformation

The kernel of T (denoted $\ker T$) and the image of T (denoted $\text{im } T$ or $T(V)$) are defined by

$$\ker T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}$$

$$\text{im } T = \{T(\mathbf{v}) \mid \mathbf{v} \in V\} = T(V)$$

The kernel of T is often called the nullspace of T because it consists of all vectors $\mathbf{v} \in V$ satisfying the condition that $T(\mathbf{v}) = \mathbf{0}$. The image of T is often called the range of T and consists of all vectors $\mathbf{w} \in W$ of the form $\mathbf{w} = T(\mathbf{v})$ for some $\mathbf{v} \in V$.

Theorem

Let $T : V \rightarrow W$ be a linear transformation.

- (i) $\ker T$ is a subspace of V
- (ii) $\text{im } T$ is a subspace of W

Definition : Nullity and Rank

Given a linear transformation $T : V \rightarrow W$:

- (i) $\dim(\ker T)$ is called the nullity of T and denoted as $\text{nullity}(T)$
- (ii) $\dim(\text{im } T)$ is called the rank of T and denoted as $\text{rank}(T)$

Definition : One-to-one and Onto Linear Transformations

Let $T : V \rightarrow W$ be a linear transformation.

- (i) T is said to be onto if $\text{im } T = W$
- (ii) T is said to be one-to-one if $T(\mathbf{v}) = T(\mathbf{v}_1)$ implies $\mathbf{v} = \mathbf{v}_1$

Theorem

If $T : V \rightarrow W$ is a linear transformation, then T is one-to-one if and only if $\ker T = \{\mathbf{0}\}$.

Note

The identity transformation $1_V : V \rightarrow V$ is both one-to-one and onto for any vector space V .

Theorem

Let A be an $m \times n$ matrix, and let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear transformation induced by A , that is $T_A(\mathbf{x}) = A\mathbf{x}$ for all columns $\mathbf{x} \in \mathbb{R}^n$.

- (i) T_A is onto if and only if $\text{rank } A = m$
- (ii) T_A is one-to-one if and only if $\text{rank } A = n$

Theorem : Dimension Theorem

Let $T : V \rightarrow W$ be any linear transformation and assume that $\ker T$ and $\text{im } T$ are both finite dimensional. Then V is also finite dimensional and

$$\dim V = \dim(\ker T) + \dim(\text{im } T)$$

In other words, $\dim V = \text{nullity}(T) + \text{rank}(T)$.

Theorem

Let $T : V \rightarrow W$ be a linear transformation, and let $\{\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ be a basis of V such that $\{\mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ is a basis of $\ker T$. Then $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_r)\}$ is a basis of $\text{im } T$, and hence $r = \text{rank } T$.

Definition : Isomorphic Vector Spaces

A linear transformation $T : V \rightarrow W$ is called an isomorphism if it is both one-to-one and onto. The vector spaces V and W are said to be isomorphic if there exists an isomorphism $T : V \rightarrow W$, and we write $V \cong W$ when this is the case.

Note

An isomorphism $T : V \rightarrow W$ induces a pairing

$$\mathbf{v} \leftrightarrow T(\mathbf{v})$$

between vectors $\mathbf{v} \in V$ and vectors $T(\mathbf{v}) \in W$ that preserves vector addition and scalar multiplication. Hence, as far as their vector space properties are concerned, the spaces V and W are identical except for notation.

Theorem

If V and W are finite dimensional spaces, the following conditions are equivalent for a linear transformation $T : V \rightarrow W$.

- (i) T is an isomorphism
- (ii) If $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is any basis of V , then $\{T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)\}$ is a basis of W .
- (iii) There exists a basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ of V such that $\{T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)\}$ is a basis of W .

Theorem

If V and W are finite dimensional vector spaces, then $V \cong W$ if and only if $\dim V = \dim W$

Corollary

Let U, V and W denote vector spaces. Then:

- (i) $V \cong V$ for every vector space V
- (ii) If $V \cong W$ then $W \cong V$
- (iii) If $U \cong V$ and $V \cong W$, then $U \cong W$

Corollary

If V is a vector space and $\dim V = n$, then V is isomorphic to $\mathbb{R}^n$.

Definition : Coordinate Isomorphism

Let V be an n -dimensional vector space with basis $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$. The coordinate isomorphism corresponding to B is the linear transformation $C_B : V \rightarrow \mathbb{R}^n$ defined by

$$C_B(v_1\mathbf{b}_1 + \dots + v_n\mathbf{b}_n) = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

It satisfies $C_B(\mathbf{b}_i) = \mathbf{e}_i$ for each i and is an isomorphism.

Theorem

If V and W have the same dimension n , a linear transformation $T : V \rightarrow W$ is an isomorphism if it is either one-to-one or onto.

Definition : Composition of Linear Transformations

Given linear transformations $V \xrightarrow{T} W \xrightarrow{S} U$, the composite $ST : V \rightarrow U$ of T and S is defined by

$$ST(\mathbf{v}) = S[T(\mathbf{v})] \quad \forall \mathbf{v} \in V$$

The operation of forming the new function ST is called composition (sometimes denoted $S \circ T$).

Theorem

Let $V \xrightarrow{T} W \xrightarrow{S} U \xrightarrow{R} Z$ be linear transformations.

- (i) The composite ST is again a linear transformation
- (ii) $T1_V = T$ and $1_W T = T$
- (iii) $(RS)T = R(ST)$

Theorem

Let V and W be finite dimensional vector spaces. The following conditions are equivalent for a linear transformation $T : V \rightarrow W$.

- (i) T is an isomorphism
- (ii) There exists a linear transformation $S : W \rightarrow V$ such that $ST = 1_V$ and $TS = 1_W$

Definition : Inverse of Linear Transformations

Given an isomorphism $T : V \rightarrow W$, there exists another isomorphism called the inverse of T , denoted by T^{-1} . $T : V \rightarrow W$ and $T^{-1} : W \rightarrow V$ are related by the fundamental identities:

$$T^{-1}[T(\mathbf{v})] = \mathbf{v} \quad \forall \mathbf{v} \in V \quad \text{and} \quad T[T^{-1}(\mathbf{w})] = \mathbf{w} \quad \forall \mathbf{w} \in W$$

Lemma : Orthogonal Lemma in $\mathbb{R}^n$

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be an orthogonal set in $\mathbb{R}^n$. Given $\mathbf{x} \in \mathbb{R}^n$, write

$$\mathbf{f}_{m+1} = \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m$$

Then:

- (i) $\mathbf{f}_{m+1} \cdot \mathbf{f}_k = 0$ for $k = 1, 2, \dots, m$
- (ii) If $\mathbf{x} \notin \text{span}\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, then $\mathbf{f}_{m+1} \neq 0$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_m, \mathbf{f}_{m+1}\}$ is an orthogonal set.

Theorem

Let U be a subspace of $\mathbb{R}^n$

- (i) Every orthogonal subset $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ in U is a subset of an orthogonal basis of U
- (ii) U has an orthogonal basis

Theorem : Gram-Schmidt Orthogonalization Algorithm in $\mathbb{R}^n$

If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ is any basis of a subspace U of $\mathbb{R}^n$, construct $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m$ in U successively as follows:

$$\begin{aligned}\mathbf{f}_1 &= \mathbf{x}_1 \\ \mathbf{f}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 \\ \mathbf{f}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 \\ &\vdots \\ \mathbf{f}_k &= \mathbf{x}_k - \frac{\mathbf{x}_k \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x}_k \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\mathbf{x}_k \cdot \mathbf{f}_{k-1}}{\|\mathbf{f}_{k-1}\|^2} \mathbf{f}_{k-1}\end{aligned}$$

for each $k = 2, 3, \dots, m$. Then:

- (i) $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is an orthogonal basis of U
- (ii) $\text{span}\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\} = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ for each $k = 1, 2, \dots, m$

Remark

Observe that the vector $\frac{\mathbf{x} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2} \mathbf{f}_i$ is unchanged if a nonzero scalar multiple of $\mathbf{f}_i$ is used in place of $\mathbf{f}_i$. Hence, if a newly constructed $\mathbf{f}_i$ is multiplied by a nonzero scalar at some stage of the Gram-Schmidt algorithm, the subsequent $\mathbf{f}$'s will be unchanged. This is useful in actual calculations.

Definition : Orthogonal Complement of a Subspace of $\mathbb{R}^n$

If U is a subspace of $\mathbb{R}^n$, define the orthogonal complement $U^\perp$ of U by:

$$U^\perp = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = 0 \ \forall \mathbf{y} \in U\}$$

Lemma

Let U be a subspace of $\mathbb{R}^n$

- (i) $U^\perp$ is a subspace of $\mathbb{R}^n$
- (ii) $\{\mathbf{0}\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{\mathbf{0}\}$
- (iii) If $U = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$, then $U^\perp = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{x}_i = 0 \text{ for } i = 1, 2, \dots, k\}$

Definition : Projection onto a Subspace of $\mathbb{R}^n$

Let U be a subspace of $\mathbb{R}^n$ with orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$. If $\mathbf{x} \in \mathbb{R}^n$, the vector

$$\text{proj}_U \mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m$$

is called the orthogonal projection of $\mathbf{x}$ on U . For the zero subspace $U = \{\mathbf{0}\}$, we define

$$\text{proj}_{\{\mathbf{0}\}} \mathbf{x} = \mathbf{0}$$

Theorem : Projection Theorem in $\mathbb{R}^n$

If U is a subspace of $\mathbb{R}^n$ and $\mathbf{x} \in \mathbb{R}^n$, write $\mathbf{p} = \text{proj}_U \mathbf{x}$. Then:

- (i) $\mathbf{p} \in U$ and $\mathbf{x} - \mathbf{p} \in U^\perp$
- (ii) $\mathbf{p}$ is the vector in U closest to $\mathbf{x}$ in the sense that

$$\|\mathbf{x} - \mathbf{p}\| < \|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{y} \in U, \mathbf{y} \neq \mathbf{p}$$

Theorem

Let U be a fixed subspace of $\mathbb{R}^n$. If we define $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by

$$T(\mathbf{x}) = \text{proj}_U \mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$

- (i) T is a linear operator
- (ii) $\text{im } T = U$ and $\ker T = U^\perp$
- (iii) $\dim U + \dim U^\perp = n$

Theorem : Orthogonal Matrix Criteria

The following conditions are equivalent for an $n \times n$ matrix P .

- (i) P is invertible and $P^{-1} = P^T$
- (ii) The rows of P are orthonormal
- (iii) The columns of P are orthonormal

Definition : Orthogonal Matrices

An $n \times n$ matrix P is called an orthogonal matrix if it satisfies one (and hence all) of the conditions in the “Orthogonal Matrix Criteria” theorem.

Definition : Orthogonal Diagonalizable Matrices

An $n \times n$ matrix A is said to be orthogonally diagonalizable when an orthogonal matrix P can be found such that $P^{-1}AP = P^TAP$ is diagonal.

Theorem : Principal Axes Theorem for Matrices

The following conditions are equivalent for an $n \times n$ matrix A .

- (i) A has an orthonormal set of n eigenvectors
- (ii) A is orthogonally diagonalizable
- (iii) A is symmetric

Since the eigenvalues of a (real) symmetric matrix are real, this theorem is sometimes called the real spectral theorem, and the set of distinct eigenvalues is called the spectrum of the matrix.

Definition : Principal Axes

A set of orthonormal eigenvectors of a symmetric matrix A is called a set of principal axes for A .

Theorem

If A is an $n \times n$ symmetric matrix, then

$$(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A\mathbf{y})$$

for all columns $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Theorem

If A is a symmetric matrix, then eigenvectors of A corresponding to distinct eigenvalues are orthogonal.

Definition : Quadratic Forms & Principal Axes

If A is symmetric and a set of orthogonal eigenvectors of A is given, the eigenvectors are called principal axes of A . An expression $q = ax_1^2 + bx_1x_2 + cx_2^2$ is called a quadratic form in the variables x_1 and x_2 , and the graph of the equation $q = 1$ is called a conic in these variables.

If we introduce new variables y_1 and y_2 by setting $x_1 = y_1 + y_2$ and $x_2 = y_1 - y_2$, then $q = y_1^2 - y_2^2$, a diagonal form with no cross term y_1y_2 . Then, the y_1 and y_2 axes are called the principal axes for the conic.

Theorem : Triangulation Theorem

If A is an $n \times n$ matrix with n real eigenvalues, an orthogonal matrix P exists such that P^TAP is upper triangular.

Corollary

If A is an $n \times n$ matrix with real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (not necessarily distinct), then $\det A = \lambda_1 \lambda_2 \dots \lambda_n$ and $\operatorname{tr} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$.

Definition : Positive Definite Matrices

A square matrix is called positive definite if it is symmetric and all its eigenvalues λ are positive.

Since these matrices are symmetric, the principal axes theorem plays a central role in the theory.

Theorem

If A is positive definite, then it is invertible and $\det A > 0$.

Theorem

A symmetric matrix A is positive definite if and only if $\mathbf{x}^T A \mathbf{x} > 0$ for every column $\mathbf{x} \neq \mathbf{0} \in \mathbb{R}^n$

Definition : Principal Submatrices

If A is any $n \times n$ matrix, let ${}^{(r)}A$ denote the $r \times r$ submatrix in the upper left corner of A ; that is, ${}^{(r)}A$ is the matrix obtained from A by deleting the last $n - r$ rows and columns. The matrices ${}^{(1)}A, {}^{(2)}A, {}^{(3)}A, \dots, {}^{(n)}A = A$ are called the principal submatrices of A .

Lemma

If A is positive definite, so is each principal submatrix ${}^{(r)}A$ for $r = 1, 2, \dots, n$.

Theorem

The following conditions are equivalent for a symmetric $n \times n$ matrix A :

- (i) A is positive definite
- (ii) $\det ({}^{(r)}A) > 0$ for each $r = 1, 2, \dots, n$
- (iii) $A = U^T U$ where U is an upper triangular matrix with positive entries on the main diagonal

The factorization in (iii) is unique, and called the Cholesky factorization of A .

Theorem : Algorithm for the Cholesky Factorization

If A is a positive definite matrix, the Cholesky factorization $A = U^T U$ can be obtained as follows:

1. Carry A to an upper triangular matrix U_1 with positive diagonal entries using row operations each of which adds a multiple of a row to a lower row
2. Obtain U from U_1 by dividing each row of U_1 by the square root of the diagonal entry in that row

Definition : Complex Matrix

A matrix $A = [a_{ij}]$ is called a complex matrix if every entry a_{ij} is a complex number.

Definition : Complex Conjugate

If $z = a + bi$ is a complex number, the conjugate of z is also a complex number, denoted by $\bar{z}$, given by:

$$\bar{z} = a - bi$$

Definition : Conjugate Matrix

If A is a complex matrix, the conjugate of $A = [a_{ij}]$ is the matrix:

$$\bar{A} = [\bar{a}_{ij}]$$

obtained from A by conjugating every entry.

The properties

$$\overline{A + B} = \bar{A} + \bar{B} \quad \text{and} \quad \overline{AB} = \bar{A} \bar{B}$$

hold for all complex matrices of appropriate sizes.

Definition : Standard Inner Product in $\mathbb{C}^n$

Given $\mathbf{z} = (z_1, z_2, \dots, z_n)$ and $\mathbf{w} = (w_1, w_2, \dots, w_n)$ in $\mathbb{C}^n$, define their standard inner product by:

$$\langle \mathbf{z}, \mathbf{w} \rangle = z_1 \bar{w}_1 + z_2 \bar{w}_2 + \dots + z_n \bar{w}_n = \mathbf{z} \cdot \bar{\mathbf{w}}$$

Theorem

Let $\mathbf{z}, \mathbf{z}_1, \mathbf{w}, \mathbf{w}_1$ denote vectors in $\mathbb{C}^n$, and let λ denote a complex number.

- (i) $\langle \mathbf{z} + \mathbf{z}_1, \mathbf{w} \rangle = \langle \mathbf{z}, \mathbf{w} \rangle + \langle \mathbf{z}_1, \mathbf{w} \rangle$ and $\langle \mathbf{z}, \mathbf{w} + \mathbf{w}_1 \rangle = \langle \mathbf{z}, \mathbf{w} \rangle + \langle \mathbf{z}, \mathbf{w}_1 \rangle$
- (ii) $\langle \lambda \mathbf{z}, \mathbf{w} \rangle = \lambda \langle \mathbf{z}, \mathbf{w} \rangle$ and $\langle \mathbf{z}, \lambda \mathbf{w} \rangle = \bar{\lambda} \langle \mathbf{z}, \mathbf{w} \rangle$
- (iii) $\langle \mathbf{z}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{z} \rangle}$
- (iv) $\langle \mathbf{z}, \mathbf{z} \rangle \geq 0$, and $\langle \mathbf{z}, \mathbf{z} \rangle = 0$ iff $\mathbf{z} = \mathbf{0}$

Definition : Norm and Length in $\mathbb{C}^n$

We define the norm or length $\|\mathbf{z}\|$ of a vector $\mathbf{z} = (z_1, z_2, \dots, z_n) \in \mathbb{C}^n$ by:

$$\|\mathbf{z}\| = \sqrt{\langle \mathbf{z}, \mathbf{z} \rangle} = \sqrt{|z_1|^2 + |z_2|^2 + \dots + |z_n|^2}$$

Theorem

If $\mathbf{z} \in \mathbb{C}^n$, then:

- (i) $\|\mathbf{z}\| \geq 0$ and $\|\mathbf{z}\| = 0$ iff $\mathbf{z} = \mathbf{0}$
- (ii) $\|\lambda \mathbf{z}\| = |\lambda| \|\mathbf{z}\| \quad \forall \lambda \in \mathbb{C}$

Definition : Conjugate Transpose in $\mathbb{C}^n$

The conjugate transpose A^H of a complex matrix A is defined by:

$$A^H = (\overline{A})^T = \overline{(A^T)}$$

Theorem

Let A and B denote complex matrices, and let λ be a complex number.

- (i) $(A^H)^H = A$
- (ii) $(A + B)^H = A^H + B^H$
- (iii) $(\lambda A)^H = \overline{\lambda} A^H$
- (iv) $(AB)^H = B^H A^H$

Definition : Hermitian Matrices

A square complex matrix A is called hermitian if $A^H = A$, or equivalently if $\overline{A} = A^T$.

Theorem

An $n \times n$ complex matrix A is hermitian if and only if:

$$\langle A\mathbf{z}, \mathbf{w} \rangle = \langle \mathbf{z}, A\mathbf{w} \rangle$$

for all n -tuples $\mathbf{z}, \mathbf{w} \in \mathbb{C}^n$.

Theorem

Let A denote a hermitian matrix.

- (i) The eigenvalues of A are real
- (ii) Eigenvectors of A corresponding to distinct eigenvalues are orthogonal

Definition : Orthogonal and Orthonormal Vectors in $\mathbb{C}^n$

As in the real case, a set of nonzero vectors $\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_m\}$ in $\mathbb{C}^n$ is called orthogonal if $\langle \mathbf{z}_i, \mathbf{z}_j \rangle = 0$ whenever $i \neq j$, and it is orthonormal if, in addition, $\|\mathbf{z}_i\| = 1$ for each i .

Theorem

The following are equivalent for an $n \times n$ complex matrix A .

- (i) A is invertible and $A^{-1} = A^H$
- (ii) The rows of A are an orthonormal set in $\mathbb{C}^n$
- (iii) The columns of A are an orthonormal set in $\mathbb{C}^n$

Definition : Unitary Matrices

A square complex matrix U is unitary if $U^{-1} = U^H$. Thus, a real matrix is unitary iff it is orthogonal.

Definition : Unitary Diagonalizable

An $n \times n$ complex matrix A is called unitarily diagonalizable if $U^H A U$ is diagonal for some unitary matrix U .

Theorem : Schur's Theorem

If A is any $n \times n$ complex matrix, there exists a unitary matrix U such that

$$U^H A U = T$$

is upper triangular. Moreover, the entries on the main diagonal of T are the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of A (including multiplicities).

Corollary

Let A be an $n \times n$ complex matrix, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ denote the eigenvalues of A , including multiplicities. Then,

$$\det A = \lambda_1 \lambda_2 \dots \lambda_n \quad \text{and} \quad \operatorname{tr} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$$

Theorem : Spectral Theorem

If A is hermitian, there is a unitary matrix U such that $U^H A U$ is diagonal.

Definition : Normal Matrix

An $n \times n$ complex matrix N is called normal if $N N^H = N^H N$.

Theorem

An $n \times n$ complex matrix A is unitarily diagonalizable if and only if A is normal.

Theorem : Cayley-Hamilton Theorem

If A is an $n \times n$ complex matrix, then $c_A(A) = 0$; that is, A is a root of its characteristic polynomial.

Definition : Quadratic Form

A quadratic form q in the n variables $x_1, x_2, \dots, x_n$ is a linear combination of terms $x_1^2, x_2^2, \dots, x_n^2$ and cross terms $x_1x_2, x_1x_3, x_2x_3, \dots$

Theorem : Diagonalization Theorem

Let $q = \mathbf{x}^T A \mathbf{x}$ be a quadratic form in the variables $x_1, x_2, \dots, x_n$, where $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ and A is a symmetric $n \times n$ matrix. Let P be an orthogonal matrix such that $P^T A P$ is diagonal, and define new variables $\mathbf{y} = (y_1, y_2, \dots, y_n)^T$ by:

$$\mathbf{x} = P\mathbf{y} \quad \Leftrightarrow \quad \mathbf{y} = P^T \mathbf{x}$$

If q is expressed in terms of these new variables $y_1, y_2, \dots, y_n$, the result is:

$$q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A repeated according to their multiplicities.

Theorem

If $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is a quadratic form given by a symmetric matrix A , then A is uniquely determined by q .

Definition : Congruent Matrices

Two $n \times n$ matrices A and B are called congruent, written $A \simeq B$ if $B = U^T A U$ for some invertible matrix U . Some properties of congruence are:

- (i) $A \simeq A$ for all A
- (ii) If $A \simeq B$, then $B \simeq A$
- (iii) If $A \simeq B$ and $B \simeq C$, then $A \simeq C$
- (iv) If $A \simeq B$, then A is symmetric iff B is symmetric
- (v) If $A \simeq B$, then $\text{rank } A = \text{rank } B$

Theorem : Sylvester's Law of Inertia

If $A \simeq B$, then A and B have the same number of positive eigenvalues, counting multiplicities.

Definition : Index

The index of a symmetric matrix A is the number of positive eigenvalues of A . If $q = q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is a quadratic form, the index and rank of q are defined to be the index and rank of the matrix A respectively.

Note : Complete Diagonalization

Let $q = q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ be any quadratic form in n variables, of index k and rank r where A is symmetric. We claim there are new variables $\mathbf{z}$ can be found so that q is completely diagonalized, that is,

$$q(\mathbf{z}) = z_1^2 + \cdots + z_k^2 - z_{k+1}^2 - \cdots - z_r^2$$

If $k \leq r \leq n$, let $D_n(k, r)$ denote the $n \times n$ diagonal matrix whose main diagonal consists of k 1's, followed by $r - k$ -1's, followed by $n - r$ 0's. Then we seek variables $\mathbf{z}$ such that:

$$q(\mathbf{z}) = \mathbf{z}^T D_n(k, r) \mathbf{z}$$

To determine $\mathbf{z}$, first diagonalize A as follows: Find an orthogonal matrix P_0 such that:

$$P_0^T A P_0 = D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_r, 0, \dots, 0)$$

is diagonal with the nonzero eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_r$ of A on the main diagonal (followed by $n - r$ zeros). By reordering the columns of P_0 , if necessary, we may assume that $\lambda_1, \dots, \lambda_k$ are positive and $\lambda_{k+1}, \dots, \lambda_r$ are negative. This being the case, let D_0 be the $n \times n$ diagonal matrix:

$$D_0 = \text{diag}\left(\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_k}}, \frac{1}{\sqrt{-\lambda_{k+1}}}, \dots, \frac{1}{\sqrt{\lambda_r}}, 1, \dots, 1\right)$$

Then $D_0^T D D_0 = D_n(k, r)$, so if new variables $\mathbf{z}$ are given by $\mathbf{x} = (P_0 D_0) \mathbf{z}$, we obtain:

$$q(\mathbf{z}) = \mathbf{z}^T D_n(k, r) \mathbf{z} = z_1^2 + \cdots + z_k^2 - z_{k+1}^2 - \cdots - z_r^2$$

as required. Note that the change-of-variables matrix $P_0 D_0$ from $\mathbf{z}$ to $\mathbf{x}$ has orthogonal columns (in fact, scalar multiples of the columns of P_0).

Theorem

Let A and B be symmetric $n \times n$ matrices, and let $0 \leq k \leq r \leq n$.

- (i) A has index k and rank r if and only if $A \simeq D_n(k, r)$
- (ii) $A \simeq B$ if and only if they have the same rank and index

Definition : Ordered Basis

An ordered basis is a basis $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ where the order of the vectors is taken into account.

Definition : Coordinate Vector $C_B(\mathbf{v})$

Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is an ordered basis of a vector space V , and let $\mathbf{v} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + \dots + v_n\mathbf{b}_n$. Then, the coordinate vector of $\mathbf{v}$ with respect to B is defined to be

$$C_B(\mathbf{v}) = (v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + \dots + v_n\mathbf{b}_n) = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Note that $C_B(\mathbf{b}_i) = \mathbf{e}_i$ is column i of I_n .

Theorem

If V has dimension n and $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is any ordered basis of V , the coordinate transformation $C_B : V \rightarrow \mathbb{R}^n$ is an isomorphism. In fact, $C_B^{-1} : \mathbb{R}^n \rightarrow V$ is given by:

$$C_B^{-1} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + \dots + v_n\mathbf{b}_n \quad \forall \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

Definition : Matrix of T corresponding to the ordered bases B and D

$M_{DB}(T)$ of $T : V \rightarrow W$ for bases B and D is called the matrix of T corresponding to the ordered bases B and D , we use the following notation:

$$M_{DB}(T) = \begin{bmatrix} C_D[T(\mathbf{b}_1)] & C_D[T(\mathbf{b}_2)] & \cdots & C_D[T(\mathbf{b}_n)] \end{bmatrix}$$

Theorem

Let $T : V \rightarrow W$ be a linear transformation where $\dim V = n$ and $\dim W = m$, and let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and D be ordered bases of V and W , respectively. Then the matrix $M_{DB}(T)$ is the unique $m \times n$ matrix A that satisfies:

$$C_D T = T_A C_B$$

Hence the defining property of $M_{DB}(T)$ is:

$$C_D [T(\mathbf{v})] = M_{DB}(T) C_B(\mathbf{v}) \quad \forall \mathbf{v} \in V$$

The matrix $M_{DB}(T)$ is given in terms of its columns by:

$$M_{DB}(T) = [C_D[T(\mathbf{b}_1)] \quad C_D[T(\mathbf{b}_2)] \quad \cdots \quad C_D[T(\mathbf{b}_n)]]$$

Theorem

Let $V \xrightarrow{T} W \xrightarrow{S} U$ be linear transformations and let B , D , and E be finite ordered bases of V , W , and U , respectively. Then:

$$M_{EB}(ST) = M_{ED}(S) \cdot M_{DB}(T)$$

Theorem

Let $T : V \rightarrow W$ be a linear transformation, where $\dim V = \dim W = n$. The following are equivalent:

- (i) $M_{DB}(T)$ is invertible for all ordered bases B and D of V and W
- (ii) $M_{DB}(T)$ is invertible for some pair of ordered bases B and D of V and W

When this is the case, $[M_{DB}(T)]^{-1} = M_{BD}(T^{-1})$.

Definition : Rank of a Linear Transformation

The rank of a linear transformation $T : V \rightarrow W$ is $\text{rank } T = \dim(\text{im } T)$. Moreover, if A is any $m \times n$ matrix and $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the matrix transformation, then $\text{rank}(T_A) = \text{rank } A$.

Theorem

Let $T : V \rightarrow W$ be a linear transformation where $\dim V = n$ and $\dim W = m$. If B and D are any ordered bases of V and W , then $\text{rank } T = \text{rank } [M_{DB}(T)]$.

Note

If $T : V \rightarrow V$ is a linear operator where $\dim V = n$, it is possible to choose bases B and D of V such that the matrix $M_{DB}(T)$ has a very simple form:

$$M_{DB}(T) = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where $r = \text{rank } T$.

Definition : Matrix $M_B(T)$ of $T : V \rightarrow V$ for basis B

If $T : V \rightarrow V$ is an operator on a vector space V , and if B is an ordered basis of V , define $M_B(T) = M_{BB}(T)$ and call this the B -matrix of T .

Recall that if $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear operator and $E = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$, then $C_B(\mathbf{x}) = \mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^n$, so $M_E(T) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}$ is the standard matrix of the operator T .

Theorem

Let $T : V \rightarrow V$ be an operator where $\dim V = n$, and let B be an ordered basis of V .

- (i) $C_B(T(\mathbf{v})) = M_B(T) C_B(\mathbf{v}) \quad \forall \mathbf{v} \in V$
- (ii) If $S : V \rightarrow V$ is another operator on V , then $M_B(ST) = M_B(S) M_B(T)$
- (iii) T is an isomorphism if and only if $M_B(T)$ is invertible. In this case $M_D(T)$ is invertible for every ordered basis D of V
- (iv) If T is an isomorphism, then $M_B(T^{-1}) = [M_B(T)]^{-1}$
- (v) If $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$, then $M_B(T) = \begin{bmatrix} C_B[T(\mathbf{b}_1)] & C_B[T(\mathbf{b}_2)] & \cdots & C_B[T(\mathbf{b}_n)] \end{bmatrix}$

Definition : Change Matrix $P_{D \leftarrow B}$ for bases B and D

Define the change matrix $P_{D \leftarrow B}$ by:

$$P_{D \leftarrow B} = M_{DB}(1_V) \quad \text{for any ordered bases } B \text{ and } D \text{ of } V$$

Theorem

Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and D denote ordered bases of a vector space V . Then the change matrix $P_{D \leftarrow B}$ is given in terms of its columns by:

$$P_{D \leftarrow B} = [C_D(\mathbf{b}_1) \ C_D(\mathbf{b}_2) \ \cdots \ C_D(\mathbf{b}_n)]$$

and has the property that:

$$C_D(\mathbf{v}) = P_{D \leftarrow B} C_B(\mathbf{v}) \quad \forall \mathbf{v} \in V$$

Moreover, if E is another ordered basis of V , we have:

- (i) $P_{B \leftarrow B} = I_n$
- (ii) $P_{D \leftarrow B}$ is invertible and $(P_{D \leftarrow B})^{-1} = P_{B \leftarrow D}$
- (iii) $P_{E \leftarrow D} P_{D \leftarrow B} = P_{E \leftarrow B}$

Theorem : Similarity Theorem

Let B_0 and B be two ordered bases of a finite dimensional vector space V . If $T : V \rightarrow V$ is any linear operator, the matrices $M_B(T)$ and $M_{B_0}(T)$ of T with respect to these bases are similar. More precisely,

$$M_B(T) = P^{-1} M_{B_0}(T) P$$

where $P = P_{B_0 \leftarrow B}$ is the change matrix from B to B_0 .

Theorem

Let A be an $n \times n$ matrix and let E be the standard basis of $\mathbb{R}^n$.

- (i) Let A' be similar to A , say $A' = P^{-1}AP$, and let B be the ordered basis of $\mathbb{R}^n$ consisting of the columns of P in order. Then $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is linear and:

$$M_E(T_A) = A \quad \text{and} \quad M_B(T_A) = A'$$

- (ii) If B is any ordered basis of $\mathbb{R}^n$, let P be the (invertible) matrix whose columns are the vectors in B in order. Then:

$$M_B(T_A) = P^{-1}AP$$

Definition : Similarity Invariant

A property of $n \times n$ matrices is similarity invariant if, whenever a given $n \times n$ matrix A which has the property, every matrix similar to A also has the property.

Definition : Determinant

If $T : V \rightarrow V$ is a linear operator on a finite dimensional space V , define the determinant of T (denoted $\det T$) by:

$$\det T = \det M_B(T), \quad B \text{ any basis of } V$$

Definition : Trace

If $T : V \rightarrow V$ is a linear operator on a finite dimensional space V , define the trace of T (denoted $\operatorname{tr} T$) by:

$$\operatorname{tr} T = \operatorname{tr} M_B(T), \quad B \text{ any basis of } V$$

Definition : Characteristic Polynomial

If $T : V \rightarrow V$ is a linear operator on a finite dimensional space V , define the characteristic polynomial of T by:

$$c_T(x) = c_A(x) \text{ where } A = M_B(T), \quad B \text{ any basis of } V$$

In other words, the characteristic polynomial of an operator T is the characteristic polynomial of any matrix representing T .

Definition : Image under T

If U is a subspace of V , write its image under T as:

$$T(U) = \{T(\mathbf{u}) \mid \mathbf{u} \in U\}$$

Definition : T -invariant Subspace

Let $T : V \rightarrow V$ be an operator. A subspace $U \subseteq V$ is called T -invariant if $T(U) \subseteq U$, that is $T(\mathbf{u}) \in U$ for every vector $\mathbf{u} \in U$. Hence, T is a linear operator on the vector space U .

Theorem

Let $T : V \rightarrow V$ be any linear operator. Then:

- (i) $\{\mathbf{0}\}$ and V are T -invariant subspaces
- (ii) Both $\ker T$ and $\operatorname{im} T = T(V)$ are T -invariant subspaces
- (iii) If U and W are T -invariant subspaces, so are $T(U)$, $U \cap W$, and $U + W$

Definition : Restriction of an Operator

Let $T : V \rightarrow V$ be a linear operator. If U is any T -invariant subspaces of V , then:

$$T : U \rightarrow U$$

is a linear operator on the subspace U , called the restriction of T to U

Theorem

Let $T : V \rightarrow V$ be a linear operator where V has dimension n and suppose that U is any T -invariant subspace of V . Let $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_k\}$ be any basis of U and extend it to a basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_k, \mathbf{b}_{k+1}, \dots, \mathbf{b}_n\}$ of V in any way. Then $M_B(T)$ has the block triangular form:

$$M_B(T) = \begin{bmatrix} M_{B_1}(T) & Y \\ 0 & Z \end{bmatrix}$$

where Z is $(n - k) \times (n - k)$ and $M_{B_1}(T)$ is the matrix of the restriction of T to U .

Theorem

Let A be a block upper triangular matrix, say

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ 0 & A_{22} & A_{23} & \cdots & A_{2n} \\ 0 & 0 & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & A_{nn} \end{bmatrix}$$

where the diagonal blocks are square. Then:

- (i) $\det A = (\det A_{11}) (\det A_{22}) (\det A_{33}) \cdots (\det A_{nn})$
- (ii) $c_A(x) = c_{A_{11}}(x) c_{A_{22}}(x) c_{A_{33}}(x) \cdots c_{A_{nn}}(x)$

Definition : One Dimensional T -Invariant Subspace

Let $T : V \rightarrow V$ be a linear operator. A one dimensional subspace $\mathbb{R}\mathbf{v}$, $\mathbf{v} \neq 0$, is T -invariant if and only if $T(r\mathbf{v}) = rT(\mathbf{v})$ lies in $\mathbb{R}\mathbf{v}$ for all $r \in \mathbb{R}$. This holds if and only if $T(\mathbf{v})$ lies in $\mathbb{R}\mathbf{v}$, that is, $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\lambda \in \mathbb{R}$.

Definition : Eigenvalue of an Operator

A real number λ is called an eigenvalue of an operator $T : V \rightarrow V$ if:

$$T(\mathbf{v}) = \lambda \mathbf{v}$$

holds for some nonzero vector $\mathbf{v} \in V$. In this case, $\mathbf{v}$ is called an eigenvector of T corresponding to λ .

Definition : Eigenspace of a Linear Operator

Let $T : V \rightarrow V$. The subspace:

$$E_\lambda(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \lambda \mathbf{v}\}$$

is called the eigenspace of T corresponding to λ .

Theorem

If A is an $n \times n$ matrix, a real number λ is an eigenvalue of the matrix operator $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ if and only if λ is an eigenvalue of the matrix A . Moreover, the eigenspaces agree:

$$E_\lambda(T_A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \lambda \mathbf{x}\} = E_\lambda(A)$$

Theorem

Let $T : V \rightarrow V$ be a linear operator where $\dim V = n$ and let B denote any ordered basis of V , and let $C_B : V \rightarrow \mathbb{R}^n$ denote the coordinate isomorphism. Then:

- (i) The eigenvalues λ of T are precisely the eigenvalues of the matrix $M_B(T)$ and thus are the roots of the characteristic polynomial $c_T(x)$
- (ii) In this case, the eigenspaces $E_\lambda(T)$ and $E_\lambda[M_B(T)]$ are isomorphic via the restriction $C_B : E_\lambda(T) \rightarrow E_\lambda[M_B(T)]$

Theorem

Each eigenspace of the linear operator $T : V \rightarrow V$ is a T -invariant subspace of V .

Definition : Direct Sum of Subspaces

A vector space V is said to be the direct sum of subspaces U and W if:

$$U \cap W = \{\mathbf{0}\} \quad \text{and} \quad U + W = V$$

In this case, we write $V = U \oplus W$. Given a subspace U , any subspace W such that $U \oplus W$ is called a complement of U in V .

Theorem

Let U and W be subspaces of a finite dimensional vector space V . The following three conditions are equivalent:

- (i) $V = U \oplus W$
- (ii) Each vector $\mathbf{v} \in V$ can be written uniquely in the form

$$\mathbf{v} = \mathbf{u} + \mathbf{w} \quad \text{where } \mathbf{u} \in U, \mathbf{w} \in W$$

- (iii) If $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ are bases of U and W respectively, then $B = \{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_m\}$ is a basis of V .

Theorem

Let $T : V \rightarrow V$ be a linear operator where V has dimension n . Suppose $V = U_1 \oplus U_2$ where both U_1 and U_2 are T -invariant. If $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_k\}$ and $B_2 = \{\mathbf{b}_{k+1}, \dots, \mathbf{b}_n\}$ are bases of U_1 and U_2 respectively, then:

$$B = \{\mathbf{b}_1, \dots, \mathbf{b}_k, \mathbf{b}_{k+1}, \dots, \mathbf{b}_n\}$$

is a basis of V , and $M_B(T)$ has the block diagonal form:

$$M_B(T) = \begin{bmatrix} M_{B_1}(T) & 0 \\ 0 & M_{B_2}(T) \end{bmatrix}$$

where $M_{B_1}(T)$ and $M_{B_2}(T)$ are the matrices of the restrictions of T to U_1 and to U_2 respectively.

Definition : Reducible Linear Operator

A linear operator $T : V \rightarrow V$ is said to be reducible if nonzero T -invariant subspaces U_1 and U_2 can be found such that $V = U_1 \oplus U_2$.

Definition : Involution

Let $T : V \rightarrow V$ be a linear operator such that $T^2 = 1_V$. Then, T is an involution.

Definition : Inner Product Spaces

An inner product on a real vector space V is a function that assigns a real number $\langle \mathbf{v}, \mathbf{w} \rangle$ to every pair $\mathbf{v}, \mathbf{w}$ of vectors in V in such a way that the following axioms are satisfied:

- (i) $\langle \mathbf{v}, \mathbf{w} \rangle$ is a real number for all $\mathbf{v}, \mathbf{w} \in V$
- (ii) $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$
- (iii) $\langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$
- (iv) $\langle r\mathbf{v}, \mathbf{w} \rangle = r\langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{v}, \mathbf{w} \in V$ and all $r \in \mathbb{R}$
- (v) $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in V

Definition : Inner Product Space

A real vector space V with an inner product $\langle \cdot, \cdot \rangle$ is called an inner product space.

Note

Every subspace of an inner product space is again an inner product space using the same inner product.

Theorem : L

Let V be an inner product space with inner product $\langle \cdot, \cdot \rangle$; let $\mathbf{v}, \mathbf{u}$, and $\mathbf{w}$ denote vectors in V , and let r denote a real number.

- (i) $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$
- (ii) $\langle \mathbf{v}, r\mathbf{w} \rangle = r\langle \mathbf{v}, \mathbf{w} \rangle = \langle r\mathbf{v}, \mathbf{w} \rangle$
- (iii) $\langle \mathbf{v}, \mathbf{0} \rangle = 0 = \langle \mathbf{0}, \mathbf{v} \rangle$
- (iv) $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$

Theorem

If A is any $n \times n$ positive definite matrix, then

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T A \mathbf{y} \quad \text{for all columns } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

defines an inner product on $\mathbb{R}^n$, and every inner product on $\mathbb{R}^n$ arises in this way.

Note

Just as every linear operator $\mathbb{R}^n \rightarrow \mathbb{R}^n$ corresponds to an $n \times n$ matrix, every inner product on $\mathbb{R}^n$ corresponds to a positive definite $n \times n$ matrix. In particular, the dot product corresponds to the identity matrix I_n .

Remark

If we refer to the inner product space $\mathbb{R}^n$ without specifying the inner product, we mean that the dot product is to be used.

Definition : Norm and Distance

As in $\mathbb{R}^n$, if $\langle \cdot, \cdot \rangle$ is an inner product on a space V , the norm $\|\mathbf{v}\|$ of a vector $\mathbf{v} \in V$ is defined by:

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$

We define the distance between vectors $\mathbf{v}$ and $\mathbf{w}$ in an inner product space to be:

$$d(\mathbf{v}, \mathbf{w}) = \|\mathbf{v} - \mathbf{w}\|$$

Definition : Unit Vector

A vector $\mathbf{v}$ in an inner product space V is called a unit vector if $\|\mathbf{v}\| = 1$. The set of all unit vectors in V is called the unit ball in V .

Theorem

If $\mathbf{v} \neq \mathbf{0}$ is any vector in an inner product space V , then $\frac{1}{\|\mathbf{v}\|}\mathbf{v}$ is the unique unit vector that is a positive multiple of $\mathbf{v}$.

Theorem : Cauchy-Schwarz Inequality

If $\mathbf{v}$ and $\mathbf{w}$ are two vectors in an inner product space V , then

$$\langle \mathbf{v}, \mathbf{w} \rangle^2 \leq \|\mathbf{v}\|^2 \|\mathbf{w}\|^2$$

Moreover, equality occurs if and only if one of $\mathbf{v}$ and $\mathbf{w}$ is a scalar multiple of the other.

Theorem

If V is an inner product, the norm $\|\cdot\|$ has the following properties:

- (i) $\|\mathbf{v}\| \geq 0$ for every vector $\mathbf{v} \in V$
- (ii) $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$
- (iii) $\|r\mathbf{v}\| = |r|\|\mathbf{v}\|$ for every $\mathbf{v} \in V$ and every $r \in \mathbb{R}$
- (iv) $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$ for all $\mathbf{v}, \mathbf{w} \in V$ (triangle inequality)

Theorem

Let V be an inner product space.

- (i) $d(\mathbf{v}, \mathbf{w}) \geq 0$ for all $\mathbf{v}, \mathbf{w} \in V$
- (ii) $d(\mathbf{v}, \mathbf{w}) = 0$ if and only if $\mathbf{v} = \mathbf{w}$
- (iii) $d(\mathbf{v}, \mathbf{w}) = d(\mathbf{w}, \mathbf{v})$ for all $\mathbf{v}, \mathbf{w} \in V$
- (iv) $d(\mathbf{v}, \mathbf{w}) \leq d(\mathbf{v}, \mathbf{u}) + d(\mathbf{u}, \mathbf{w})$ for all $\mathbf{v}, \mathbf{u}, \mathbf{w} \in V$

Definition : Orthogonality

Two vectors $\mathbf{v}$ and $\mathbf{w}$ in an inner product space V are said to be orthogonal if

$$\langle \mathbf{v}, \mathbf{w} \rangle = 0$$

Definition : Orthogonal and Orthonormal Sets

A set $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ of vectors is called an orthogonal set of vectors if:

- (i) Each $\mathbf{f}_i \neq \mathbf{0}$
- (ii) $\langle \mathbf{f}_i, \mathbf{f}_j \rangle = 0$ for all $i \neq j$

If, in addition, $\|\mathbf{f}_i\| = 1$ for each i , the set $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ is called an orthonormal set.

Theorem : Pythagoras' Theorem

If $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ is an orthogonal set of vectors, then:

$$\|\mathbf{f}_1 + \mathbf{f}_2 + \dots + \mathbf{f}_n\|^2 = \|\mathbf{f}_1\|^2 + \|\mathbf{f}_2\|^2 + \dots + \|\mathbf{f}_n\|^2$$

Theorem

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ be an orthogonal set of vectors.

- (i) $\{r_1\mathbf{f}_1, r_2\mathbf{f}_2, \dots, r_n\mathbf{f}_n\}$ is also orthogonal for any $r_i \neq 0 \in \mathbb{R}$
- (ii) $\left\{ \frac{1}{\|\mathbf{f}_1\|}\mathbf{f}_1, \frac{1}{\|\mathbf{f}_2\|}\mathbf{f}_2, \dots, \frac{1}{\|\mathbf{f}_n\|}\mathbf{f}_n \right\}$ is an orthonormal set

The process of passing from an orthogonal set to an orthonormal one is called normalizing the orthogonal set.

Theorem

Every orthogonal set of vectors is linearly independent.

Theorem : Expansion Theorem

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ be an orthogonal basis of an inner product space V . If $\mathbf{v}$ is any vector in V , then:

$$\mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{f}_1 \rangle}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\langle \mathbf{v}, \mathbf{f}_2 \rangle}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\langle \mathbf{v}, \mathbf{f}_n \rangle}{\|\mathbf{f}_n\|^2} \mathbf{f}_n$$

is the expansion of $\mathbf{v}$ as a linear combination of the basis vectors.

The coefficients $\frac{\langle \mathbf{v}, \mathbf{f}_1 \rangle}{\|\mathbf{f}_1\|^2} \mathbf{f}_1, \frac{\langle \mathbf{v}, \mathbf{f}_2 \rangle}{\|\mathbf{f}_2\|^2} \mathbf{f}_2, \dots, \frac{\langle \mathbf{v}, \mathbf{f}_n \rangle}{\|\mathbf{f}_n\|^2} \mathbf{f}_n$ in the expansion theorem are sometimes called the Fourier coefficients of $\mathbf{v}$ with respect to the orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$.

Definition : Lagrange Polynomials

Let $a_0, a_1, \dots, a_n$ be distinct numbers. The Lagrange polynomial corresponding to a_k is

$$\delta_k(x) = \frac{\prod_{i \neq k} (x - a_i)}{\prod_{i \neq k} (a_k - a_i)} \quad k = 0, 1, 2, \dots, n$$

It satisfies the interpolation property

$$\delta_k(a_i) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$$

with the inner product

$$\langle p(x), q(x) \rangle = \sum_{i=0}^n p(a_i) q(a_i)$$

the set $\{\delta_0, \delta_1, \dots, \delta_n\}$ forms an orthonormal basis for $\mathbf{P}_n$.

Definition : Lagrange Interpolation Expansion

Every polynomial $p(x) \in \mathbf{P}_n$ can be written as

$$p(x) = \sum_{k=0}^n p(a_k) \delta_k(x)$$

This is called the Lagrange interpolation expansion (or Lagrange interpolation formula). It reconstructs a polynomial from its values at the distinct points $a_0, a_1, \dots, a_n$.

Lemma : Orthogonal Lemma

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be an orthogonal set of vectors in an inner product space V , and let $\mathbf{v}$ be any vector not in $\text{span}\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$. Define

$$\mathbf{f}_{m+1} = \mathbf{v} - \frac{\langle \mathbf{v}, \mathbf{f}_1 \rangle}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\langle \mathbf{v}, \mathbf{f}_2 \rangle}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\langle \mathbf{v}, \mathbf{f}_m \rangle}{\|\mathbf{f}_m\|^2} \mathbf{f}_m$$

Then $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m, \mathbf{f}_{m+1}\}$ is an orthogonal set of vectors.

Theorem : Gram-Schmidt Orthogonalization Algorithm

Let V be an inner product space and let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be any basis of V . Define vectors $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n \in V$ successively as follows:

$$\begin{aligned} \mathbf{f}_1 &= \mathbf{v}_1 \\ \mathbf{f}_2 &= \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{f}_1 \rangle}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 \\ \mathbf{f}_3 &= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{f}_1 \rangle}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\langle \mathbf{v}_3, \mathbf{f}_2 \rangle}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 \\ &\vdots \\ \mathbf{f}_k &= \mathbf{v}_k - \frac{\langle \mathbf{v}_k, \mathbf{f}_1 \rangle}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\langle \mathbf{v}_k, \mathbf{f}_2 \rangle}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\langle \mathbf{v}_k, \mathbf{f}_{k-1} \rangle}{\|\mathbf{f}_{k-1}\|^2} \mathbf{f}_{k-1} \end{aligned}$$

for each $k = 2, 3, \dots, n$. Then:

- (i) $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ is an orthogonal basis of V
- (ii) $\text{span}\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ for each $k = 1, 2, \dots, n$

The purpose of the Gram-Schmidt algorithm is to convert a basis of an inner product space into an orthogonal basis. In particular, it shows that every finite dimensional inner product space has an orthogonal basis.

Corollary

If V is any n -dimensional inner product space, then V is isomorphic to $\mathbb{R}^n$ as inner product spaces. More precisely, if E is any orthonormal basis of V , the coordinate isomorphism

$$C_E : V \rightarrow \mathbb{R}^n \text{ satisfies } \langle \mathbf{v}, \mathbf{w} \rangle = C_E(\mathbf{v}) \cdot C_E(\mathbf{w})$$

for all $\mathbf{v}, \mathbf{w} \in V$.

Definition : Orthogonal Complement

Let U be a subspace of an inner product space V . As in $\mathbb{R}^n$, the orthogonal complement $U^\perp$ of U in V is defined by:

$$U^\perp = \{\mathbf{v} \in V \mid \langle \mathbf{v}, \mathbf{u} \rangle = 0 \ \forall \mathbf{u} \in U\}$$

Theorem

Let U be a finite dimensional subspace of an inner product space V

- (i) $U^\perp$ is a subspace of V and $V = U \oplus U^\perp$
- (ii) If $\dim V = n$, then $\dim U + \dim U^\perp = n$
- (iii) If $\dim V = n$, then $U^{\perp\perp} = U$

Definition : Orthogonal Projection on a Subspace

The projection on U with kernel $U^\perp$ is called the orthogonal projection on U (or simply the projection on U) and is denoted $\text{proj}_U : V \rightarrow V$

Theorem : Projection Theorem

Let U be a finite dimensional subspace of an inner product space V and let $\mathbf{v} \in V$

- (i) $\text{proj}_U : V \rightarrow V$ is a linear operator with image U and kernel $U^\perp$
- (ii) $\text{proj}_U \mathbf{v} \in U$ and $(\mathbf{v} - \text{proj}_U \mathbf{v}) \in U^\perp$
- (iii) If $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is any orthogonal basis of U , then

$$\text{proj}_U \mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{f}_1 \rangle}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\langle \mathbf{v}, \mathbf{f}_2 \rangle}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\langle \mathbf{v}, \mathbf{f}_m \rangle}{\|\mathbf{f}_m\|^2} \mathbf{f}_m$$

Note that there is no requirement that V is finite dimensional.

Theorem : Approximation Theorem

Let U be a finite dimensional subspace of an inner product space V . If $\mathbf{v}$ is any vector in V , then $\text{proj}_U \mathbf{v}$ is the vector in U that is closest to $\mathbf{v}$. Here closest means that

$$\|\mathbf{v} - \text{proj}_U \mathbf{v}\| < \|\mathbf{v} - \mathbf{u}\|$$

for all $\mathbf{u} \in U, \mathbf{u} \neq \text{proj}_U \mathbf{v}$.

Theorem

Let $T : V \rightarrow V$ be a linear operator on a finite dimensional space V . Then the following conditions are equivalent:

- (i) V has a basis consisting of eigenvectors of T
- (ii) There exists a basis B of V such that $M_B(T)$ is diagonal

Definition : Diagonalizable Linear Operators

A linear operator T on a finite dimensional space V is called diagonalizable if V has a basis consisting of eigenvectors of T .

Theorem

Let $T : V \rightarrow V$ be a linear operator on an inner product space V . If $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is an orthogonal basis of V , then

$$M_B(T) = \left[\frac{\langle \mathbf{b}_i, T(\mathbf{b}_j) \rangle}{\|\mathbf{b}_i\|^2} \right]$$

Theorem

Let V be a finite dimensional inner product space. The following conditions are equivalent for a linear operator $T : V \rightarrow V$.

- (i) $\langle \mathbf{v}, T(\mathbf{v}) \rangle = \langle T(\mathbf{v}), \mathbf{w} \rangle$ for all $\mathbf{v}, \mathbf{w} \in V$
- (ii) The matrix of T is symmetric with respect to every orthonormal basis of V
- (iii) The matrix of T is symmetric with respect to some orthonormal basis of V
- (iv) There is an orthonormal basis $B = \{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ of V such that $\langle \mathbf{f}_i, T(\mathbf{f}_j) \rangle = \langle T(\mathbf{f}_i), \mathbf{f}_j \rangle$ holds for all i and j .

Definition : Symmetric Linear Operator

A linear operator T on an inner product space V is called symmetric if $\langle \mathbf{v}, T(\mathbf{w}) \rangle = \langle T(\mathbf{v}), \mathbf{w} \rangle$ holds for all $\mathbf{v}, \mathbf{w} \in V$.

Theorem

A symmetric linear operator on a finite dimensional inner product space has real eigenvalues.

Theorem

Let $T : V \rightarrow V$ be a symmetric linear operator on an inner product space V , and let U be a T -invariant subspace of V . Then:

- (i) The restriction of T to U is a symmetric linear operator on U
- (ii) $U^\perp$ is also T -invariant

Theorem : Principal Axes Theorem

The following conditions are equivalent for a linear operator T on a finite dimensional inner product space V .

- (i) T is symmetric
- (ii) V has an orthogonal basis consisting of eigenvectors of T

Definition : Distance Preserving

If V is an inner product space, a transformation $S : V \rightarrow V$ (not necessarily linear) is said to be distance preserving if:

$$\|S(\mathbf{v}) - S(\mathbf{w})\| = \|\mathbf{v} - \mathbf{w}\| \quad \text{for all } \mathbf{v}, \mathbf{w} \in V$$

Lemma

Let V be an inner product space of dimension n , and consider a distance preserving transformation $S : V \rightarrow V$. If $S(\mathbf{0}) = \mathbf{0}$, then S is linear.

Definition : Isometries

Distance preserving linear operators are called isometries.

Theorem

If V is a finite dimensional inner product space, then every distance preserving transformation $S : V \rightarrow V$ is the composite of a translation and an isometry.

Theorem

Let $T : V \rightarrow V$ be a linear operator on a finite dimensional inner product space V . The following conditions are equivalent:

- (i) T is an isometry (T preserves distance)
- (ii) $\|T(\mathbf{v})\| = \|\mathbf{v}\| \quad \forall \mathbf{v} \in V$ (T preserves norms)
- (iii) $\langle T(\mathbf{v}), T(\mathbf{w}) \rangle = \langle \mathbf{v}, \mathbf{w} \rangle \quad \forall \mathbf{v}, \mathbf{w} \in V$ (T preserves inner products)
- (iv) If $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$ is an orthonormal basis of V , then $\{T(\mathbf{f}_1), T(\mathbf{f}_2), \dots, T(\mathbf{f}_n)\}$ is also an orthonormal basis (T preserves orthonormal bases)
- (v) T carries some orthonormal basis to an orthonormal basis

Corollary

Let V be a finite dimensional inner product space.

- (i) Every isometry of a finite dimensional V is an isomorphism
 - (a) $1_V : V \rightarrow V$ is an isometry
 - (b) The composite of two isometries of V is an isometry
 - (c) The inverse of an isometry of V is an isometry

The conditions in (ii) assert that the set of isometries of a finite dimensional inner product space form a group.

Theorem

Let $T : V \rightarrow V$ be an operator where V is a finite dimensional inner product space. The following conditions are equivalent.

- (i) T is an isometry
- (ii) $M_B(T)$ is an orthogonal matrix for every orthonormal basis B
- (iii) $M_B(T)$ is an orthogonal matrix for some orthonormal basis B

Corollary

If $T : V \rightarrow V$ is an isometry where V is a finite dimensional inner product space, then $\det T = \pm 1$.

Theorem

Let $T : V \rightarrow V$ be an isometry on the two-dimensional inner product space V . Then there are two possibilities:

- (i) There is an orthonormal basis B of V such that

$$M_B(T) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, 0 \leq \theta < 2\pi$$

- (ii) There is an orthonormal basis B of V such that

$$M_B(T) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Furthermore, type (i) occurs if and only if $\det T = 1$, and type (ii) occurs if and only if $\det T = -1$.

Corollary

An operator $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is an isometry if and only if T is a rotation or a reflection.

Lemma

Let $T : V \rightarrow V$ be an isometry of a finite dimensional inner product space V . If U is a T -invariant subspace of V , then $U^\perp$ is also T -invariant.

Lemma

Let $T : V \rightarrow V$ be an isometry of the finite dimensional inner product space V . If λ is a complex eigenvalue of T , then $|\lambda| = 1$.

Lemma

Let $T : V \rightarrow V$ be an isometry of the n -dimensional inner product space V . If T has a non-real eigenvalue, then V has a two-dimensional T -invariant subspace.

Theorem

Let $T : V \rightarrow V$ be an isometry of the n -dimensional inner product space V . Given an angle θ , write $R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$. Then there exists an orthonormal basis B of V such that $M_B(T)$ has one of the following block diagonal forms, classified for convenience by whether n is odd or even:

$$n = 2k + 1 : \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & R(\theta_1) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R(\theta_k) \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} -1 & 0 & \cdots & 0 \\ 0 & R(\theta_1) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R(\theta_k) \end{bmatrix}$$

$$n = 2k : \begin{bmatrix} R(\theta_1) & 0 & \cdots & 0 \\ 0 & R(\theta_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R(\theta_k) \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} -1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & R(\theta_1) & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & R(\theta_{k-1}) \end{bmatrix}$$

Theorem

If $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an isometry, there are three possibilities.

(i) T is a rotation, and $M_B(T) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$ for some orthonormal basis B .

(ii) T is a reflection, and $M_B(T) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some orthonormal basis B .

(iii) $T = QR = RQ$ where Q is a reflection, R is a rotation about an axis perpendicular to the fixed plane of Q and $M_B(T) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$ for some orthonormal basis B .

Hence, T is a rotation if and only if $\det T = 1$.

Definition : Hyperplane

Let V be an n -dimensional inner product space. A subspace of V of dimension $n - 1$ is called a hyperplane in V .

Definition : Fixed Hyperplanes and Reflection

Let V be an n -dimensional inner product space, and let $Q : V \rightarrow V$ be an isometry with matrix

$$M_B(Q) = \begin{bmatrix} -1 & 0 \\ 0 & I_{n-1} \end{bmatrix}$$

for some orthonormal basis $B = \{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$. Then $Q(\mathbf{f}_1) = -\mathbf{f}_1$ whereas $Q(\mathbf{u}) = \mathbf{u}$ for each $\mathbf{u} \in U = \text{span}\{\mathbf{f}_2, \dots, \mathbf{f}_n\}$. Hence U is called the fixed hyperplane of Q and Q is called reflection in U . Note that each hyperplane in V is the fixed hyperplane of a (unique) reflection of V .

Theorem

Let $T : V \rightarrow V$ be an isometry of a finite dimensional inner product space V . Then there exist isometries $T_1, \dots, T_k$ such that

$$T = T_k T_{k-1} \cdots T_2 T_1$$

where each T_i is either a rotation or a reflection, at most one is a reflection, and $T_i T_j = T_j T_i$ holds for all i and j . Furthermore, T is a composite of rotations if and only if $\det T = 1$.